

In this section, you will learn how to use **Kiro** in a spec-driven development style. Unlike vibe coding where you give high-level instructions and let Kiro figure everything out, spec coding involves creating detailed specifications that break down complex features into manageable, well-defined tasks. This approach provides better control, documentation, and iterative development for complex projects.

Kiro's Spec feature allows you to formalize the design and implementation process by iterating on requirements, design, and implementation tasks with structured documentation before letting the agent work through the implementation.

What You'll Learn

This workshop is structured around creating comprehensive specifications for different aspects of your application:

- **Infrastructure Specifications:** Detailed AWS resource definitions, security policies, and deployment strategies
- **Frontend Specifications:** UI/UX design, component architecture, and technical implementation details
- **Backend Specifications:** API design, data models, and integration patterns
- **Security Specifications:** Authentication, authorization, encryption, and compliance requirements
- **Advanced Kiro Features:** Agent Hooks, MCP Servers, and Agent Steering for enhanced development workflows

Workshop Structure

The workshop is organized into specification-focused exercises that build upon each other:

1. **Foundation Specifications** - Requirements, architecture, and planning
2. **Infrastructure Specifications** - AWS resources, security, and deployment
3. **Frontend Specifications** - UI/UX design and technical implementation
4. **Backend Specifications** - API design and integration patterns
5. **Advanced Features** - Enhanced capabilities and Kiro-specific features

By the end, you'll have a fully functional chatbot with comprehensive specifications that serve as living documentation for your project.

The (focused) time needed to complete all the exercises is 8 to 12 hours.